

GENERATIVE AI FOR WORK & RESEARCH

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Title: Analysis the relationship between qualification and salary

Table of content

I.	Introduction	1
II.	Years of Experience Analysis	1
III.	Programming Language Analysis	2
IV.	Reflection: A short reflection	3
V.	Add Your Reflection	Erreur ! Signet non défini.

I. Introduction

This project analyses a sample of Data Analyst job postings to explore how required experience and data language skills relate to salary.

The dataset includes job information such as job title, company, location, salary estimate, minimum years of experience, and required data languages.

The main goal is to understand whether jobs requiring more experience tend to offer higher median salaries.

The project also compares median salary estimates across data language categories: R, Python, Both, and Neither.

II. Years of Experience Analysis

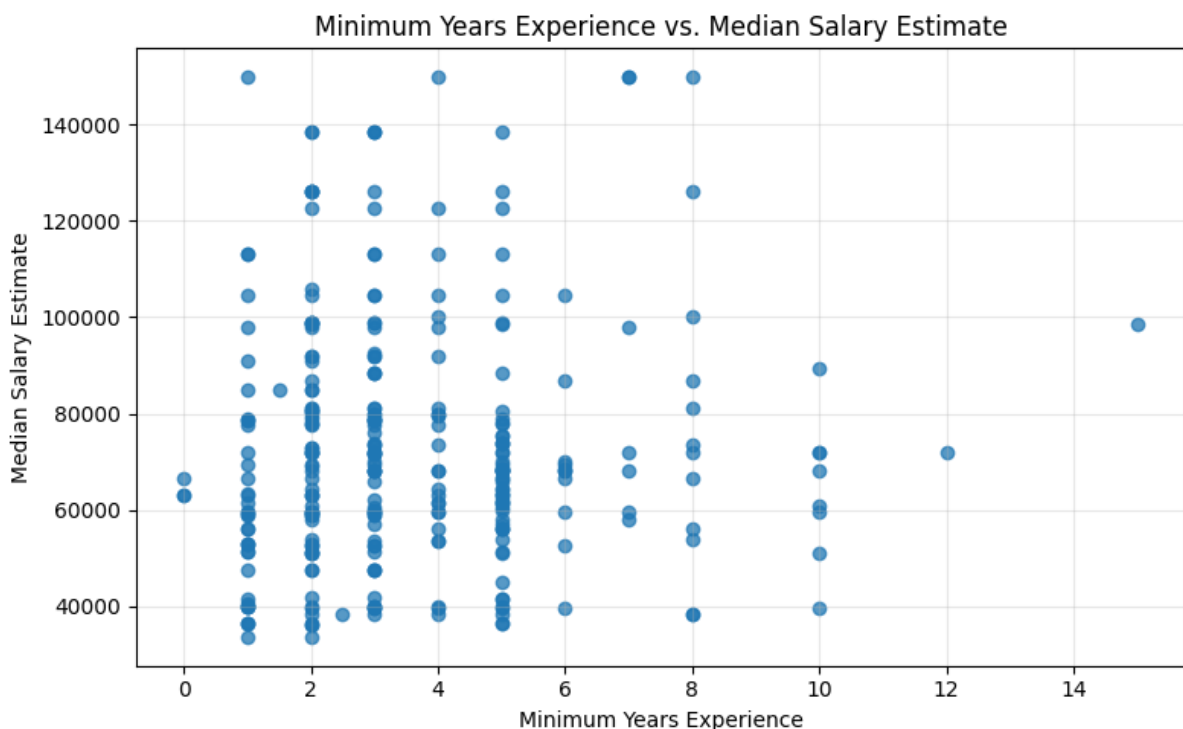
The first graph is a scatter plot showing the relationship between minimum years of experience and median salary estimate.

Overall, the graph suggests a slight positive relationship: jobs requiring more years of experience often have higher salary estimates. However, the pattern is not very strong because salaries vary widely at nearly every experience level.

Most job postings are concentrated between 1 and 5 years of required experience, with salaries commonly falling around the \$50,000 to \$90,000 range.

There are also some high salary outliers, especially around 4 to 8 years of experience, where some jobs offer salaries above \$120,000.

In summary, experience appears to influence salary, but it is not the only factor. Other variables such as location, company size, industry, and required technical skills may also affect salary.



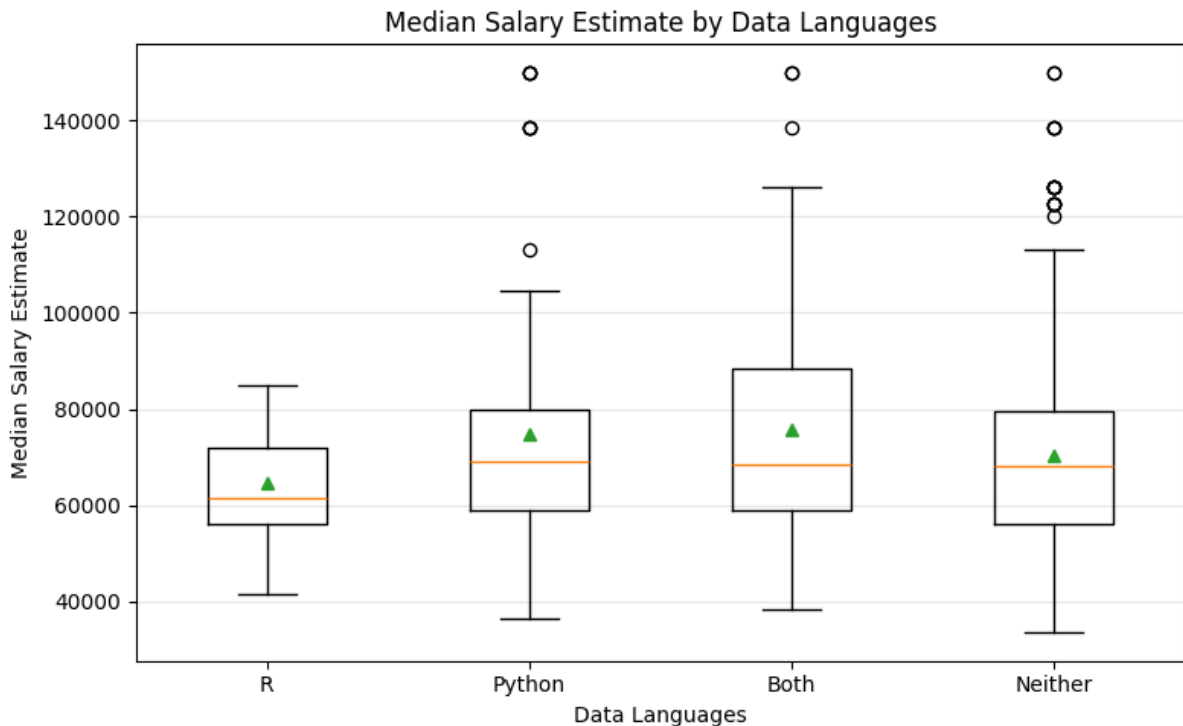
III. Programming Language Analysis

The second graph is a box plot comparing median salary estimates by data language category: R, Python, Both, and Neither.

Overall, the salaries are fairly similar across the groups, but there are some differences. Jobs requiring Python or Both Python and R appear to have slightly higher salary ranges than jobs requiring only R.

The Both category shows a wider spread of salaries, meaning jobs asking for both Python and R have more variation in pay. This could suggest that some of these roles are more advanced or specialized.

The Neither category has many observations and also includes several high salary outliers. This means some jobs may not explicitly mention Python or R but still offer high salaries, possibly because they require other tools or skills.



In summary, knowing Python, or both Python and R, may be associated with somewhat higher salaries, but the differences are not dramatic. Other factors such as experience, location, company, and industry likely also affect salary.

IV. Reflection: A short reflection

The part that I found most difficult was the process itself. There are many important steps that must be followed before using formulas in Google Sheets. I realized that it is necessary to be very rigorous when working with different languages. For example, when I copied and pasted a prompt into Google Sheets, it did not work because the formula was written in English while my Google Sheets settings were in French.

Finally, I also realized that some tasks and manipulations are actually easy to implement without using AI.